An Eye Movement Classification Method Based on Cascade Forest

Can Wang, Ruimin Wang, Yue Leng, Keiji Iramina, *Member, IEEE*, Yuankui Yang, and Sheng Ge, *Member, IEEE*

***Abstract*—Eye tracking technology has become increasingly important in scientific research and practical applications. In the field of eye tracking research, analysis of eye movement data is crucial, particularly for classifying raw eye movement data into eye movement events. Current classification methods exhibit considerable variation in adaptability across different participants, and it is necessary to address the issues of class imbalance and data scarcity in eye movement classification. In the current study, we introduce a novel eye movement classification method based on cascade forest (EMCCF), which comprises two modules: 1) a feature extraction module that employs a multi-scale time window method to extract features from raw eye movement data; 2) a classification module that innovatively employs a layered ensemble architecture, integrating the cascade forest structure with ensemble learning principles, specifically for eye movement classification. Consequently, EMCCF not only enhanced the accuracy and efficiency of eye movement classification but also represents an advancement in applying ensemble learning techniques within this domain. Furthermore, experimental results indicated that EMCCF outperformed existing deep learning-based classification models in several metrics and demonstrated robust performance across different datasets and participants.**

***Index Terms*—Cascade forest, ensemble learning, eye movement classification.**

I. Introduction

EYE tracking technology, involving analysis of the distribution of visual attention by monitoring and recording eye movements, is increasingly used in several research fields, including psychology, neuroscience, healthcare, and human-computer interaction [1], [2], [3], [4]. The wide range of applications of eye tracking technology highlights its importance as a powerful tool for scientific research and practical applications.

In eye tracking, classifying raw eye movement data into eye movement events can significantly reduce the difficulty of performing eye movement analysis [4]. These events represent eye movement patterns in different contexts [5], and four eye movement events (fixation, saccade, smooth pursuit, and blink) are commonly studied [4], [6]. These eye movement events are the target classes for eye movement classification. Existing classification methods are mainly based on manually adjusted thresholds or criteria [6], [7]. Threshold-based methods provide a practical and efficient approach for eye movement classification. However, the process of determining thresholds is characterized by significant subjectivity, caused by the wide variety of definitions and lack of a universally accepted standard for what constitutes optimal threshold levels [5], [8], [9]. Moreover, the thresholds for eye movement-related analysis are significantly influenced by the research context, objectives, and the eye-tracking technology employed [10]. For these reasons, the establishment of a uniform objective standard for optimal thresholds is complex. Moreover, the threshold is static after setting, which makes the threshold less adaptable to different scenarios and participants [8]. With the development and wide application of deep learning (DL) techniques, DL methods have been effectively introduced into the field of eye movement classification. DL-based models have been demonstrated in several studies to achieve significant improvements in classification accuracy and robustness compared with traditional algorithms [11], [12], [13]. Startsev et al. proposed a convolution neural network-bidirectional long short term memory (CNN-BLSTM) model for eye movement classification [11], which achieved very good results and substantially improved classification accuracy for smooth pursuit compared with that in previous studies. Elmadjian et al. simplified the CNN-BLSTM model to construct a convolution neural network-long short term memory (CNN-LSTM) model to achieve faster classification [12]. Elmadjian et al. used a temporal convolutional networks (TCN) model to further improve the classification accuracy of this approach [13].

Although DL-based classification methods can achieve better classification performance, all of these approaches have certain limitations. Neural networks require large datasets for training [14], leading to a data scarcity problem when applied to small-scale datasets [15]. Additionally, in eye movement

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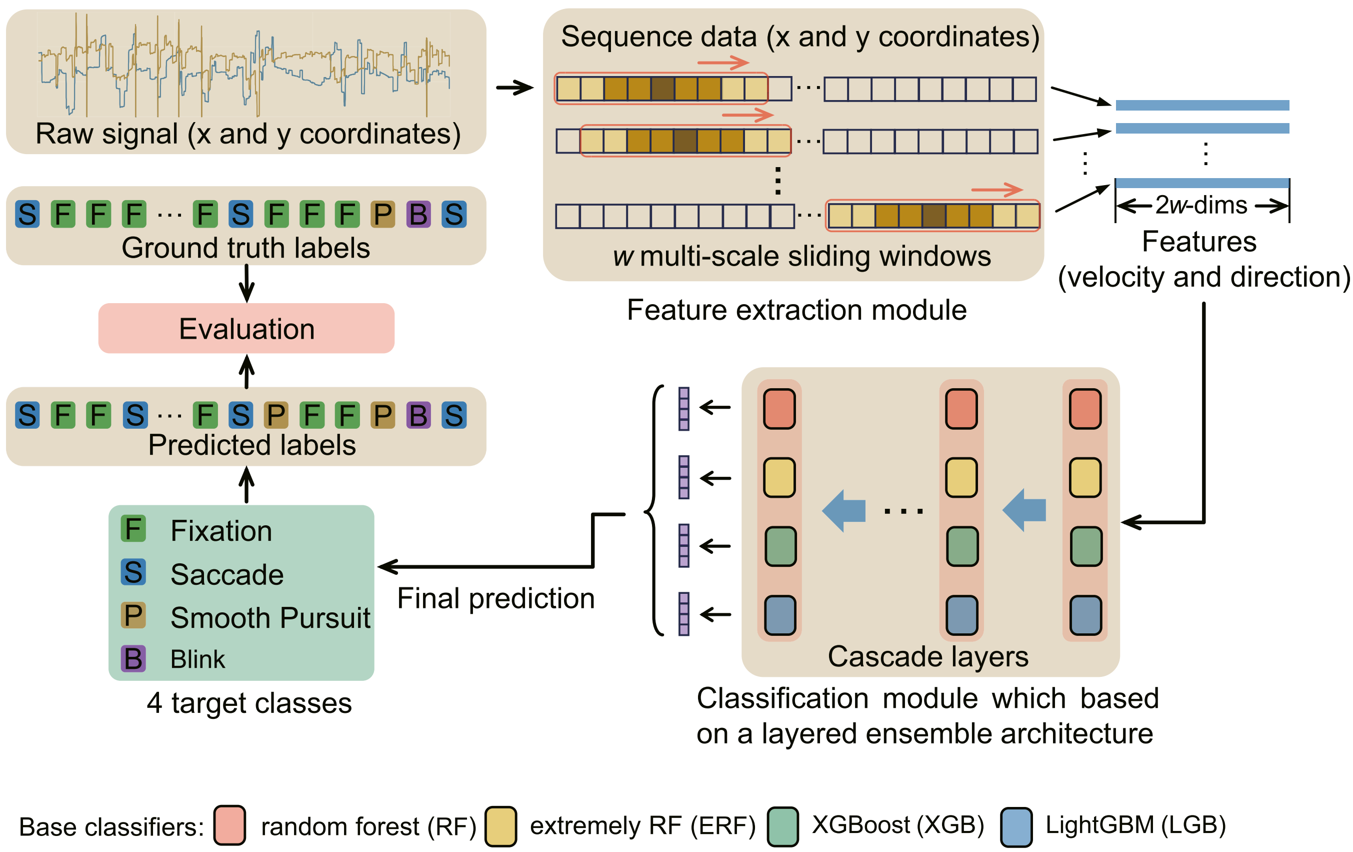

Fig. 1. Overview of the proposed method's structure. Raw data are processed through a feature extraction module followed by a classification module to achieve the final eye movement classification.

datasets, the predominance of fixation samples and the relative scarcity of other classes, such as smooth pursuit and blink, can result in a class imbalance problem [16], [17]. This imbalance adversely affects the performance of existing DL-based eye movement classification models, particularly in classifying smooth pursuit and blink [11], [12], [13]. Overall, employing DL models for eye movement classification introduces problems in both data scarcity and class imbalance, rendering DL models inadequate for eye movement classification.

To address the existing problems of DL for eye movement classification, the current study aimed to develop a comprehensive method that integrates both feature extraction and classification processes into a unified framework. Initially, we extracted features on the basis of the temporal dependence of eye movement signals using the multi-scale time window method. We then employed a layered ensemble architecture for the classification of the extracted features. This dual-module approach is referred to as the eye movement classification method based on cascade forest (EMCCF) in this study.

II. Methods and Materials

The overall framework of the EMCCF consists of two main modules: (1) a feature extraction module which utilizes multi-scale time window method to extract features from raw eye movement data; (2) a classification module that is based on a layered ensemble architecture, combining the cascade forest (CF) structure with the principles of ensemble learning. The raw eye movement signal data analyzed in this study were derived from the GazeCom dataset [18] and the head-mounted raw eye-movement dataset (HMR) [12]. The overall framework of the EMCCF is shown in Fig. 1.

TABLE I
Sample Proportions in Two Datasets

Classes	GazeCom	HMR
Fixation	72.55%	56.30%
Saccade	10.53%	6.46%
Smooth Pursuit	11.02%	25.17%
Blink	5.90%	11.77%

A. Dataset Description

Two eye movement datasets, GazeCom [18] and HMR [12], were used in this study. The proportions of eye movement data events in these two datasets are shown in Table I.

1) GazeCom Dataset: When compiling the GazeCom dataset, 54 participants were involved, each being presented with a series of 18 video clips, with each clip lasting 21 seconds. Four distinct eye movement classes were identified in this dataset: fixations, saccades, smooth pursuits, and blinks. These classes were determined on the basis of consensus of two experts who

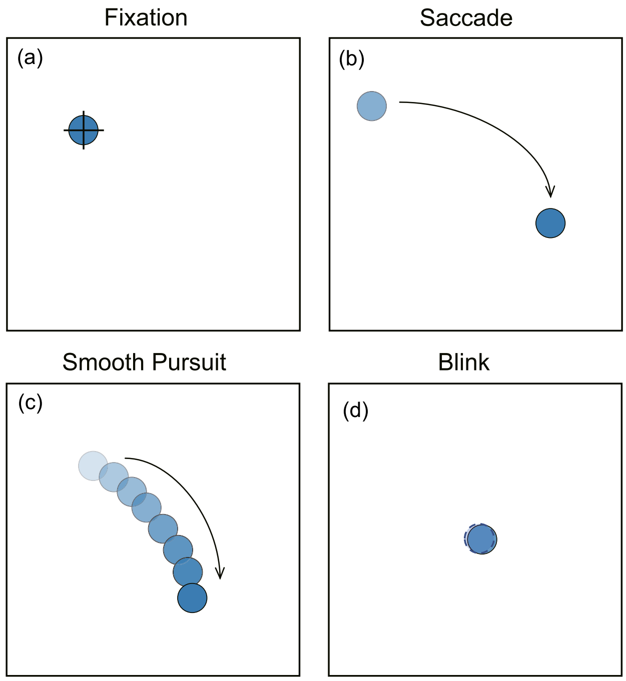

Fig. 2. Four distinct eye movement events classified in datasets. (a) Fixation: eyes remain steadily focused on stationary targets. (b) Saccade: rapid eye movements that shift focus from one fixation point to another. (c) Smooth pursuit: eyes smoothly follow a target moving at a constant speed. and (d) Blink: rapid closing and reopening of the eyelids.

reviewed the eye movement data. Data for GazeCom were collected using an eye tracker that operates at 250 Hz. The dataset comprised over 4.3 million data points. The data had a high confidence level, with an average value of 99.4% [11], [13], [18].

2) HMR Dataset: The HMR dataset was compiled from data collected from 13 participants (six women) aged between 21 and 46 years. All participants possessed either normal vision or corrected-to-normal vision. As detailed by Kassner et al. [19], a head-mounted pupil core eye tracker with a tracking rate of 200 Hz was used. During data acquisition, a 3×3 uniformly distributed nine-dot grid paradigm was employed to generate dot stimuli. Participants were required to gaze at these stimuli as they appeared. The location and color of the stimuli changed throughout the process, enabling the capture of fixation, saccade, smooth pursuit, and blink (Fig. 2). The captured data included normalized pupil center coordinates from both the left and right eyes in the eye camera view. The development of the HMR dataset was primarily motivated by two factors, i.e., the absence of datasets containing raw, unlabeled eye movement patterns captured via head-mounted trackers, and the limited representation of smooth pursuit movement samples in the GazeCom dataset.

The two datasets described above were able to effectively represent the application scenarios of eye tracking. The HMR dataset was acquired using head-mounted trackers, providing a dynamic interaction scenario of eye tracking on the basis of immersive environments (e.g., augmented reality and virtual reality). The GazeCom dataset contained more eye movement behaviors of participants while watching video content, providing an interaction scenario in a daily environment. These two datasets were chosen not only to test the adaptability and accuracy of the eye movement classification method, but also to support research in a broader range of areas, such as human visual attention research and computer vision.

B. Feature Extraction

The raw eye movement data contained only the x and y coordinates of each sample, and feature extraction from the raw data is necessary before performing eye movement classification. Considering the eye movement signals as time-series data, neighboring data points are temporally dependent on each other. Thus, this temporal dependence of data points must be considered in the feature extraction stage. In addition, the temporal dependence of the eye movement sequence is complex, and includes both long-term and short-term dependencies [20]. The temporal convolutional network (TCN) [21] excels in capturing short-term dependencies, whereas the long short-term memory network (LSTM) [22] is adept at identifying both short-term and long-term temporal relationships. These two models effectively leverage the temporal dependencies in eye movement sequences through distinct approaches, resulting in enhanced performance and accuracy in classification. To capture the temporal characteristics of eye movement signals, this study designed a new multi-scale time window method to capture and analyze eye movement data. The specific details of the multi-scale time window method are shown in Fig. 3. In eye movement data analysis, multi-scale time windows allow the model to capture eye movement patterns within various time windows; shorter time windows are more suitable for capturing saccades, whereas longer time windows are more suitable for recognizing smooth pursuit [6]. Multi-scale time window methods are used to capture eye movement signals from a time series by surrounding the current analyzed sample, extracting multiple subseries of different scales from the time series. In this way, it is possible to analyze not only the information of the sample itself but also its relationships with preceding and succeeding samples across various temporal scales. In the current study, the use of a multi-scale time window method for feature extraction aims to improve the ability of the classification algorithm to recognize complex eye movements, thus optimizing the accuracy and efficiency of eye movement classification. The use of a multi-scale time window method can fully extract the feature information of the sequences on the time series. The multi-scale time window method effectively extracts the features of different scales of the eye movement sequences. Moreover, it reduces reliance on single-scale analysis, thereby improving the accuracy of eye movement classification. In detail, the number of samples n_i contained in each sliding time window is determined by (1), and the strides set S of consisting of multi-scale windows is defined by (2):

$$n_i = 1 + 2^i \tag{1}$$

$$S = \{n_1, n_2, n_3, \ldots, n_w\} \tag{2}$$

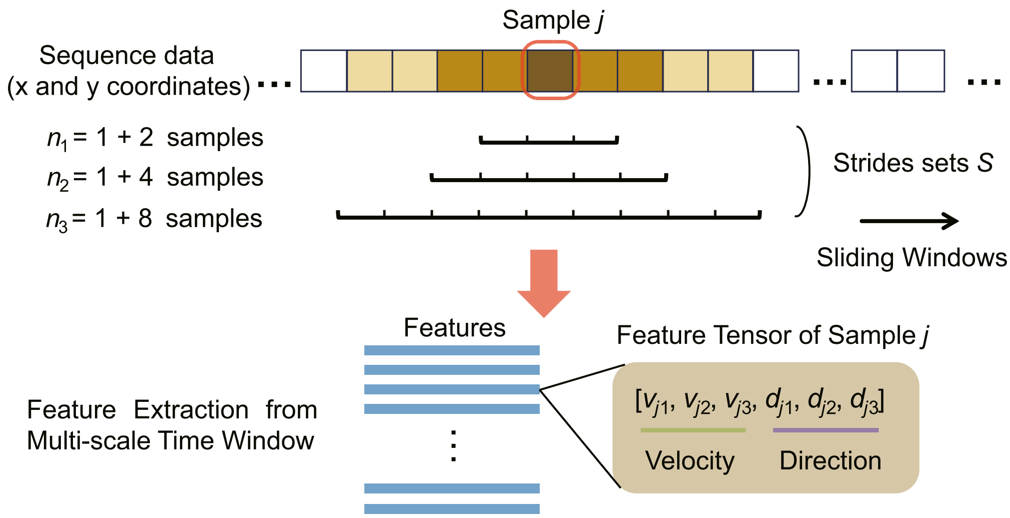

Fig. 3. Schematic of the multi-scale time window method for feature extraction. This illustration demonstrates the process using three distinct multi-scale windows for clarity. The method uses multiple sliding windows of varying scales to capture temporal information from samples adjacent to the current one. Within each window, velocity and direction features are extracted, advancing in one-step increments.

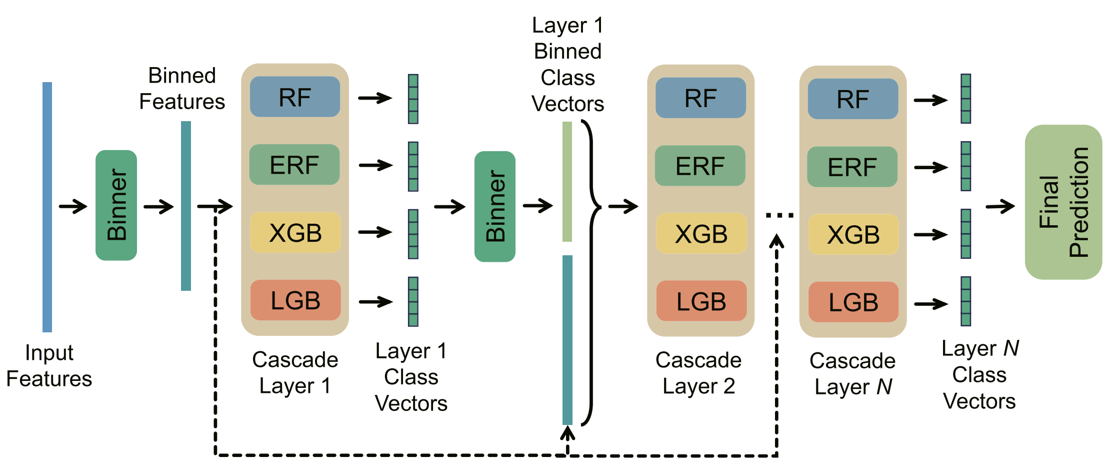

Fig. 4. The schematic illustrates the architecture of the EMCCF model. The model consists of four distinct types of base classifiers in each layer: random forest (RF), extremely randomized forest (ERF), XGBoost (XGB), and LightGBM (LGB).

where w is an integer denoting the number of windows, and for each window i there is $1 \leqslant i \leqslant w$. According to pre-analysis in Appendix A, the maximum number of multi-scale time windows is best set to 7, which can combine both operational efficiency and accuracy.

Startsev et al. [11] reported that velocity and direction were the two most effective features for classification. Velocity and direction are crucial metrics in eye movement analysis. Velocity helps distinguish between fixations, which have lower velocities, and other eye movements, which typically show higher velocities [6]. However, direction indicating the angle of eye movement, is essential for identifying saccades, which involve rapid directional changes, and smooth pursuits, characterized by smoother directional shifts [11]. Therefore, in the current study, we focused on analyzing these two features within each sliding window. The feature extraction process is shown in Fig. 3.

C. Classification Module Based on a Layered Ensemble Architecture

The classification module utilizes a layered ensemble architecture [23], employing a cascade forest (CF) structure and principles of ensemble learning (see Fig. 4). The classification module is designed around a layered ensemble architecture that incorporates a cascade forest (CF) structure, adhering to the principle of ensemble learning. Ensemble learning itself is predicated on the concept of synthesizing and integrating a multitude of base learners to execute tasks collaboratively and more effectively [24]. The application of the principle of ensemble learning in the CF structure in the classification module is primarily manifested in the following ways: First, the EMCCF model conducts a layer-by-layer sequential process, refining and transforming features on the basis of the output of the previous layer [14]. Second, each layer combines different base classifiers [25]. This combination process is reported to greatly improve classification accuracy, exceeding the performance of using a single base classifier [26]. Moreover, to increase diversity among classifiers in ensemble learning for better performance, four different base classifiers are employed in the model. These classifiers are random forest (RF), extremely randomized forest (ERF), XGBoost (XGB), and LightGBM (LGB), each containing 100 trees.

After the raw data pass through the feature extraction module, they first undergo a "Binner" process. This step can be regarded as a form of feature discretization, which transforms continuous features into discrete bins to facilitate more efficient processing by subsequent forests. The discretized features are then fed into the first cascade layer (refer to Cascade Layer 1 in Fig. 4).

In the EMCCF model, on the basis of the layer-by-layer CF structure, each cascade layer combines four different base classifiers: 1) Random forest (RF): enhances prediction accuracy and stability by building multiple decision trees and aggregating their results through majority voting [26], [27]. 2) Extremely randomized forest (ERF): similar in base structure and core idea to RF, but introduces a higher degree of randomness in tree training. This method randomly selects features and split points at nodes, further reducing model variance [27]. 3) XGBoost (XGB): a gradient boosting ensemble method that sequentially constructs trees, focusing at each step on reducing residuals left by the previous step to improve prediction precision [26]. 4) LightGBM (LGB): an efficient gradient boosting framework that utilizes histogram-based optimizations and gradient-based one-sided sampling, enhancing efficiency in processing large-scale data and reducing memory consumption [27]. The combination of four base classifiers can significantly enhance diversity among classifiers, which can significantly enhance the performance and stability of the ensemble learning model [14], [27].

Base classifiers employ different ensemble methods. RF and ERF use the bagging ensemble method, whereas XGB and LGB use the boosting ensemble method [27]. The bagging ensemble method independently trains each tree in the base classifier, while the boosting ensemble method sequentially trains each tree in the base classifier, with each tree attempting to correct the errors of the previous one [27]. Thus, the diversity among base classifiers in each layer is significantly enhanced, benefiting the model's overall performance [28]. After receiving input binned features, each cascade layer generates the class vector based on four classes and four base classifiers. Moreover, in the EMCCF model, the number of layers (N) is determined using an adaptive approach. Specifically, during the model's training process, an evaluation of the model's classification accuracy is conducted after each addition of a new layer. If no improvement in overall accuracy is observed after adding a new layer, further addition of layers is terminated, thus determining the number of layers (N).

D. Performance Evaluation

Performance evaluation was conducted by examining three aspects. First, the proposed EMCCF model was compared with existing DL-based models such as CNN-BLSTM [11], CNN-LSTM [12], and TCN [13], aiming to verify the superiority of EMCCF in classification performance compared with DL models. Second, to evaluate the impact of different base classifier combination methods on the performance of the EMCCF model, we compared four typical base classifier combination methods, confirming the effectiveness of the combination method used in EMCCF for enhancing model performance. Third, we investigated the effect of the difference in the number of multi-scale time windows employed in the feature extraction process on the classification performance of the model. Examining these three aspects enabled comprehensive evaluation of the effectiveness and applicability of the proposed EMCCF model in the eye movement classification task.

In addition to describing the specific procedure for performance evaluation, it is essential to provide a detailed explanation of the experimental configuration, including the selection of baselines, the metrics employed, and the methodologies for training and evaluation.

1) Baseline Models: TCN [13], CNN-LSTM [12], and CNN-BLSTM [11] are utilized as baseline models, representing the state-of-the-art in eye movement classification. Following the guidelines in [11] and [12], all baseline models employed a dropout rate of 0.25 and a kernel size of 5.

2) Metric of Model Performance: Eye movement classification tasks suffer from the class imbalance problem [11], [12], which leads to the classifier having low accuracy for minority classes and high accuracy for the majority class. In such cases, relying solely on accuracy as the metric may be misleading. F1 score is the harmonic mean of Precision and Recall, considering both the precision and comprehensiveness of the model [29]. Therefore, F1 score can provide a more comprehensive and objective understanding of the model's performance. For this reason, in this study, F1 score was used as the primary metric for evaluating model performance.

3) Training and Evaluation: Two datasets, Two datasets (HMR and GazeCom) were used for eye movement analysis in this study. Referencing Zhou's study [14], we allocated 80% of the data as the training set, 10% as the validation set, and the remaining 10% as the test set. The training set was used for the growth of the cascade, while the validation set was employed to verify the model's accuracy. All training and evaluation processes were conducted on a computer equipped with an Intel(R) Xeon(R) Gold 6240 CPU, 128 GB RAM, and an NVIDIA GeForce RTX 3080 GPU, running the Ubuntu 20.04.5 LTS operating system.

III. Results

A. Comparison With Baseline Models

The performance of the EMCCF model was compared with that of the baseline model. To ensure fairness and accuracy of the comparison, the data underwent the same processing method (the number of multi-scale windows was uniformly set to 7) before they were input into the EMCCF model and the baseline model (the feature extraction module). Precision, recall, and F1 scores for each classification task were calculated for all models (shown in Fig. 5).

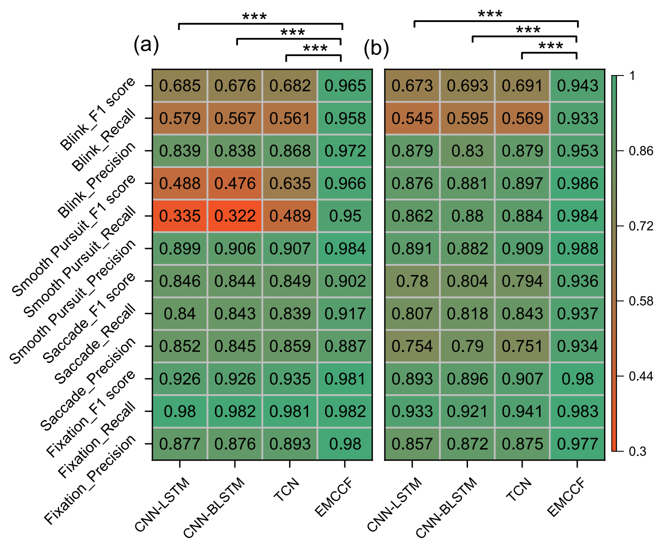

Fig. 5. The performance of eye movement classification in the EMCCF model compared to DL-based methods on (a) GazeCom dataset and (b) HMR dataset.

As illustrated in Fig. 5, the EMCCF model exhibited an improvement over the baseline models for identifying fixations, applicable to both the HMR and GazeCom datasets. Specifically, in the analysis of the HMR dataset, the EMCCF model outperformed the baseline models in terms of F1 scores for four classes: fixation, saccade, smooth pursuit, and blink. The EMCCF model exhibited a notable enhancement in the saccade class, in which its F1 score was approximately 16% higher than that of the best-performing DL model (CNN-BLSTM), and an improvement of more than 35% in the blink class. In the analysis using the GazeCom dataset, EMCCF showed slight improvements in accuracy for fixation and saccade classes and significant improvements in smooth pursuit and blink classes. Furthermore, The current study independently evaluated the performance of four classification models on two distinct datasets using one-way analysis of variance (ANOVA). For the HMR dataset, the ANOVA results revealed significant differences in classification accuracy among the models $(F[3, 204] = 14.859, p < 0.001)$. A similar significant result was observed in the analysis of the GazeCom dataset $(F[3, 860] = 84.820, p < 0.001)$. In Fig. 5, the comparison of F1 score between the EMCCF model and other models is represented by $***$, indicating that the p-value of Bonferroni post-hoc test result is less than 0.001. Overall, the EMCCF model demonstrated the most robust performance across all classes. Although the CNN-LSTM and CNN-BLSTM models tended to increase precision, they compromised on recall. The TCN model, although superior in classification accuracy compared with the other two DL models, did not reach the overall performance of the EMCCF model.

B. Inter-Participants Comparison

To evaluate model performance across participants to reveal the model's adaptability to individual differences and generalization ability, this study also analyzed the variance of the classification accuracy across participants. Analysis on the basis of the HMR dataset showed that there were no significant differences in classification accuracy among different participants for all models, except in the blink class. In the blink class, the baseline models showed greater variability in classification accuracy across participants (standard deviation: 0.026 for EMCCF, 0.249 for TCN, 0.246 for CNN-LSTM, and 0.229 for CNN-BLSTM, respectively). For the GazeCom dataset, the models showed small differences in performance across participants in the fixation and saccade classes (all standard deviations in the order of 10e-2 and below), but in the smooth pursuit and blink classes, baseline models showed significant differences (standard deviations in the smooth pursuit class: EMCCF 0.012, TCN 0.091, CNN-LSTM 0.083, and CNN-BLSTM 0.030, respectively; standard deviations in the blink class: EMCCF 0.017, TCN 0.173, and CNN-LSTM 0.174, CNN-BLSTM 0.170). The results shown in Fig. 6 indicate that the EMCCF model exhibited more robust performance than the baseline DL models.

C. Comparison of Different Ensemble Learning Approach

To investigate the impact of different base classifier combination methods on the classification performance in the EMCCF model, we conducted an analysis using four typical base classifier combination methods. These methods included: 1) two RF and two ERF, representing a bagging ensemble approach; 2) two XGB and two LGB, representing a boosting ensemble method; 3) four LGB, representing a boosting ensemble approach, but with lower diversity among the base classifiers compared with the second method; 4) EMCCF, which consisting of RF, ERF, XGB, and LGB, combines both bagging and boosting ensemble methods. The base classifier combination method in the EMCCF model showed a performance enhancement compared with the other methods (see Fig. 7). The statistical methodology employed is consistent with that in Fig. 5, involving the independent evaluation of four typical base classifier combination methods on two distinct datasets using one-way ANOVA. For the HMR dataset, significant differences were found among the models $(F[3, 204] = 27.922,\ p < 0.001)$. Similarly, significant differences were observed for the GazeCom dataset $(F[3, 860] = 249.425,\ p < 0.001)$. In Fig. 7, comparisons between the combination method of the EMCCF model and other base classifier combination methods are all marked with $***$, indicating $p < 0.001$. The significant differences shown in Fig. 7 were calculated on the basis of the F1 score. The results shown in Fig. 7 indicated that the base classifier combination method of combining bagging and boosting ensemble methods applied in the EMCCF model increases diversity and yields an increase in accuracy compared with the use of either bagging or boosting methods independently.

To comprehensively evaluate the classification performance of the EMCCF model, we used t-SNE technology to produce a visualization (see Fig. 8). This visualization compares the EMCCF model with the TCN model, which is the model that performs the best in terms of classification among the baseline DL models. The visualization shows that samples of the same

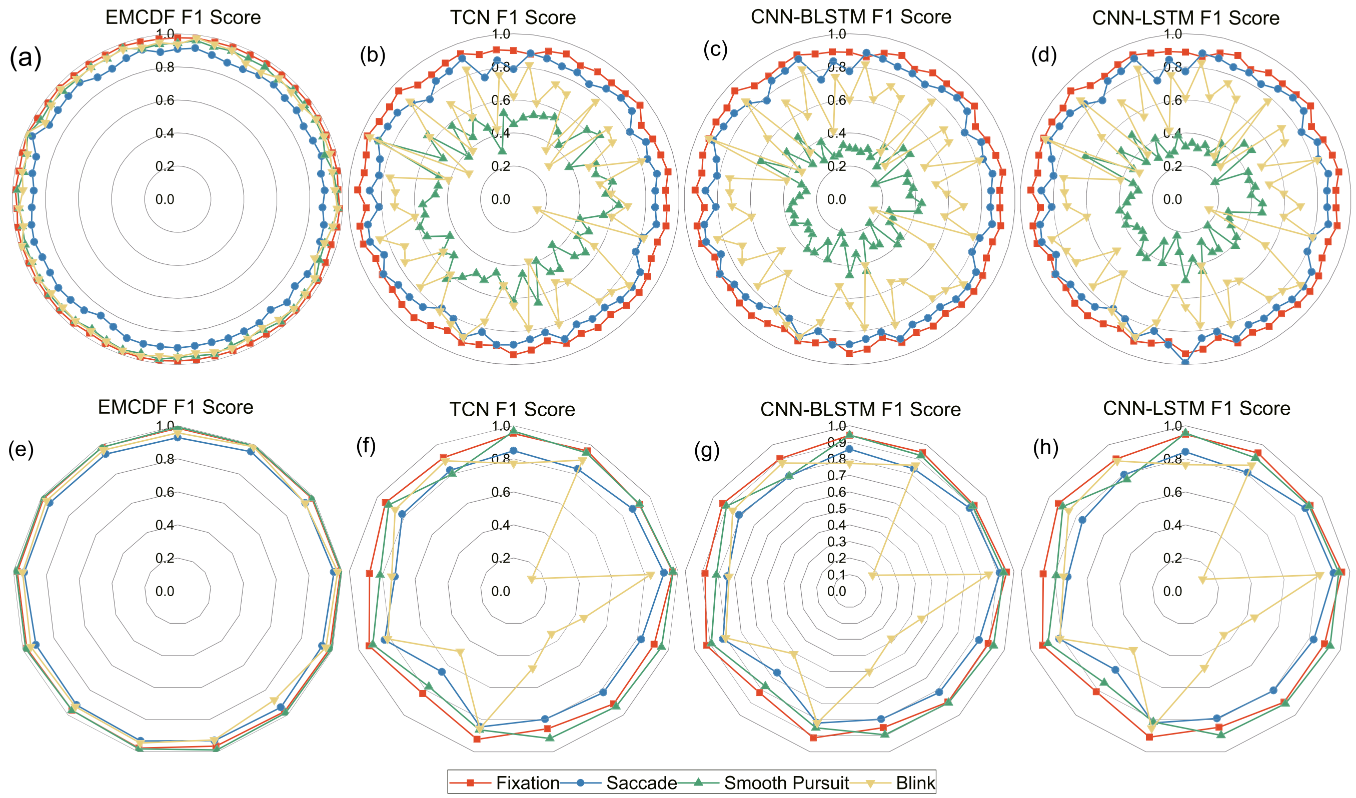

Fig. 6. Classification accuracy of each model across different participants, evaluated using the F1 score as the metric. Results based on the GazeCom dataset are presented in (a)–(d), whereas (e)–(h) show results derived from the HMR dataset.

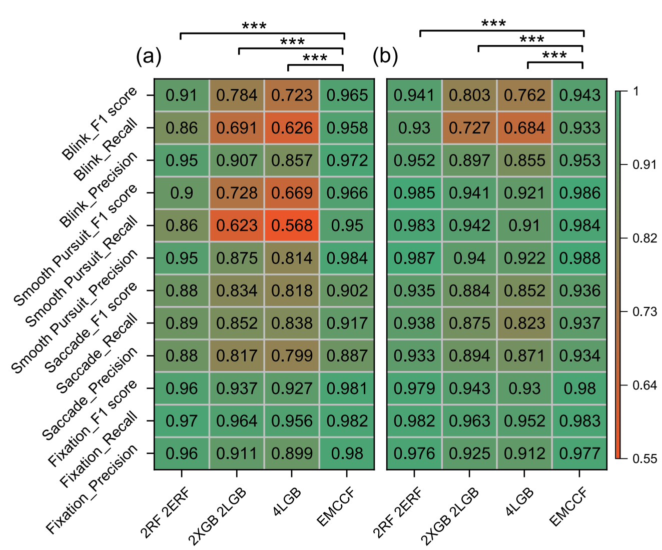

Fig. 7. Comparison of eye movement classification accuracy among different base classifier combination methods on the (a) GazeCom dataset and (b) HMR dataset.

class are more tightly clustered in the EMCCF model, forming a clear demarcation from the points of the other classes. This intuitively demonstrated the effectiveness of the EMCCF model in classifying the raw data.

D. Comparison of Different Multi-Scale Time Window Conditions

Finally, the effect of different multi-scale window conditions on the model's classification performance was investigated (Fig. 9). Compared with other DL-based models, the classification accuracy of the EMCCF model exhibited superior performance in almost all multi-scale window conditions all four classes. Moreover, the EMCCF model achieved a substantial level of classification accuracy with a limited number of multi-scale windows. Specifically, in the saccade class, the EMCCF method reached a level that was close to the best classification performance when the number of windows was 2 (the mean F1 score based on the GazeCom dataset was 0.720 and the mean F1 score based on the HMR dataset was 0.781 when the number of windows was 2), whereas the DL-based models were affected by the data scarcity problem and class imbalance problem, and could not perform effective classification.

E. Model Running Time During the Training and Testing Phases

To provide insights into the computational complexity and scalability of the proposed method, we conducted an assessment of model running time. The running time during the training and testing phases of four models were compared under identical hardware conditions, including the time required for feature extraction, as illustrated in Fig. 10. According to Fig. 10, it

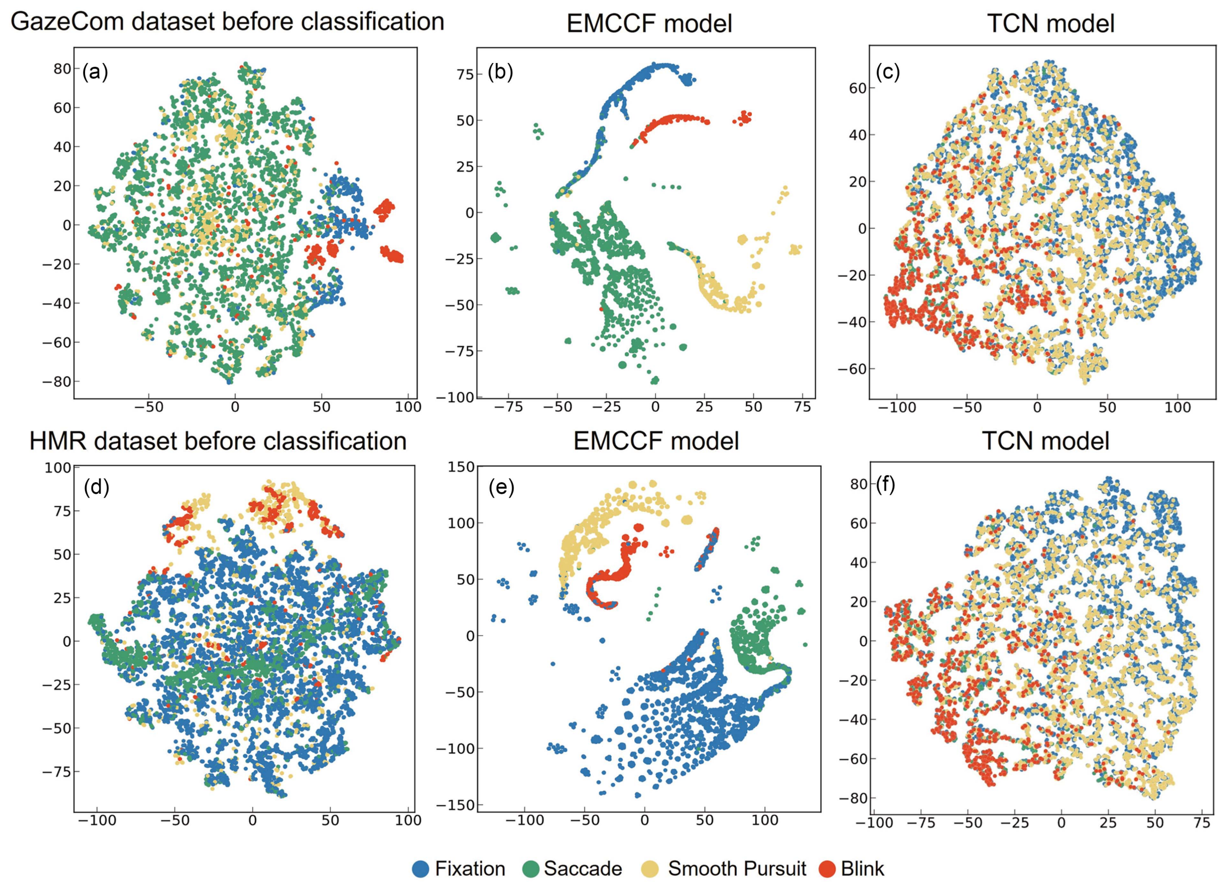

Fig. 8. t-SNE visualization of eye movement classification for selected participants from the GazeCom and HMR datasets. (a)–(c) Relate to the GazeCom dataset for participant AAF. Specifically, (a) illustrates the data prior to classification, (b) shows the classification results based on the EMCCF model, and (c) represents the outcomes using the TCN model. (d)–(f) Pertain to the HMR dataset for participant 2. Specifically, (d) shows the data before classification, (e) presents the classification results using the EMCCF model, and (f) displays the results using the TCN model.

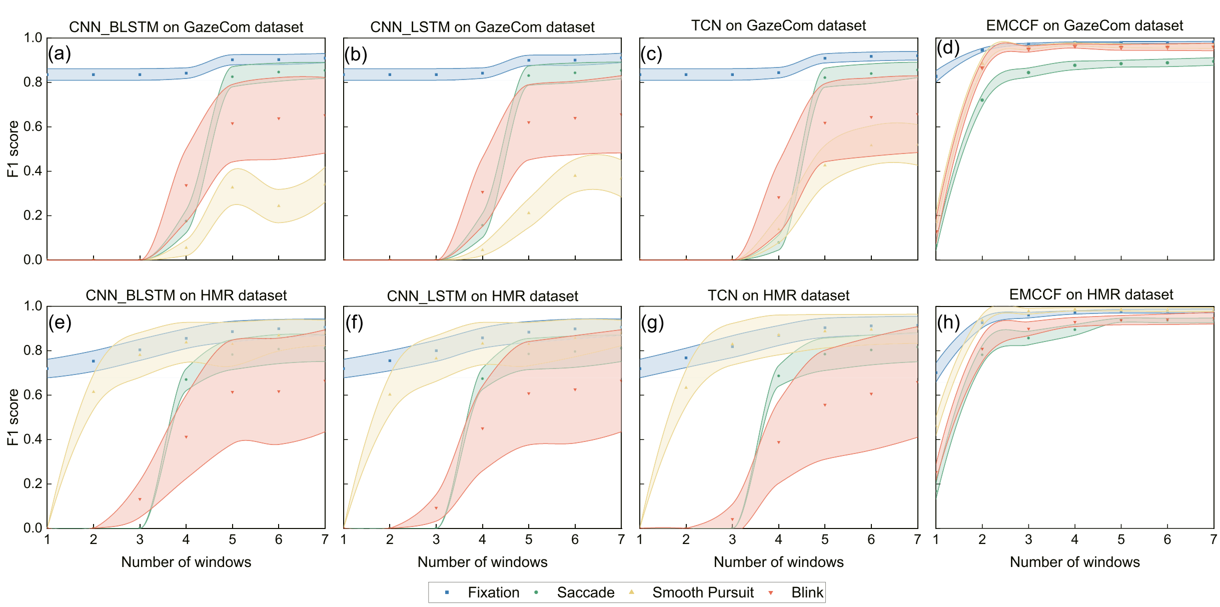

Fig. 9. Classification accuracy of each model under different multi-scale time window conditions (using F1 score as evaluation index). (a)–(d) Pertain to the GazeCom dataset and (e)–(h) relate to the HMR dataset. The shaded areas in the figure denote error bars, which represent the standard deviation of different experimental results under the same number of windows.

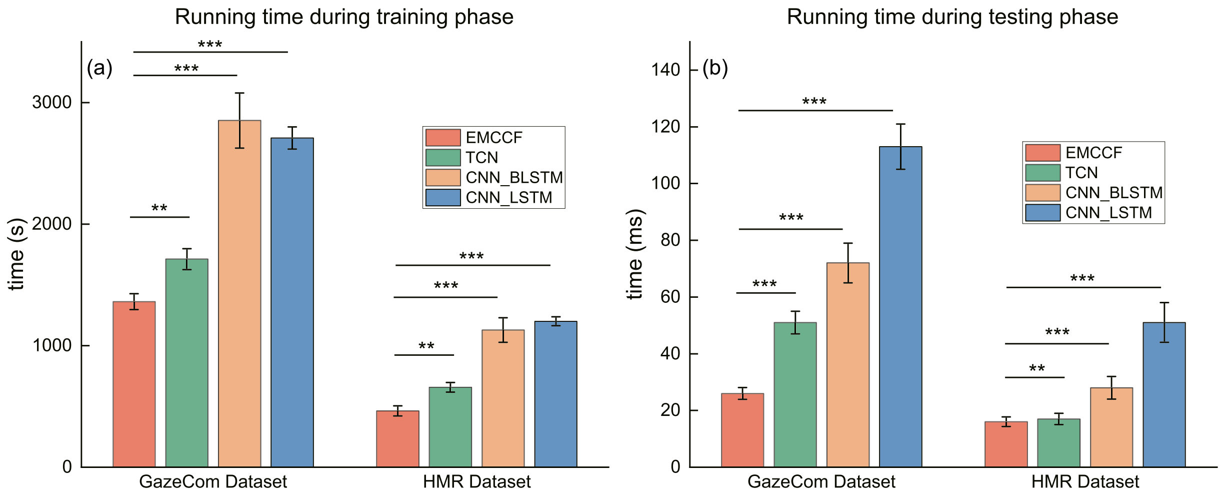

Fig. 10. Comparisons of running time among different models in (a) training phase and (b) testing phase.

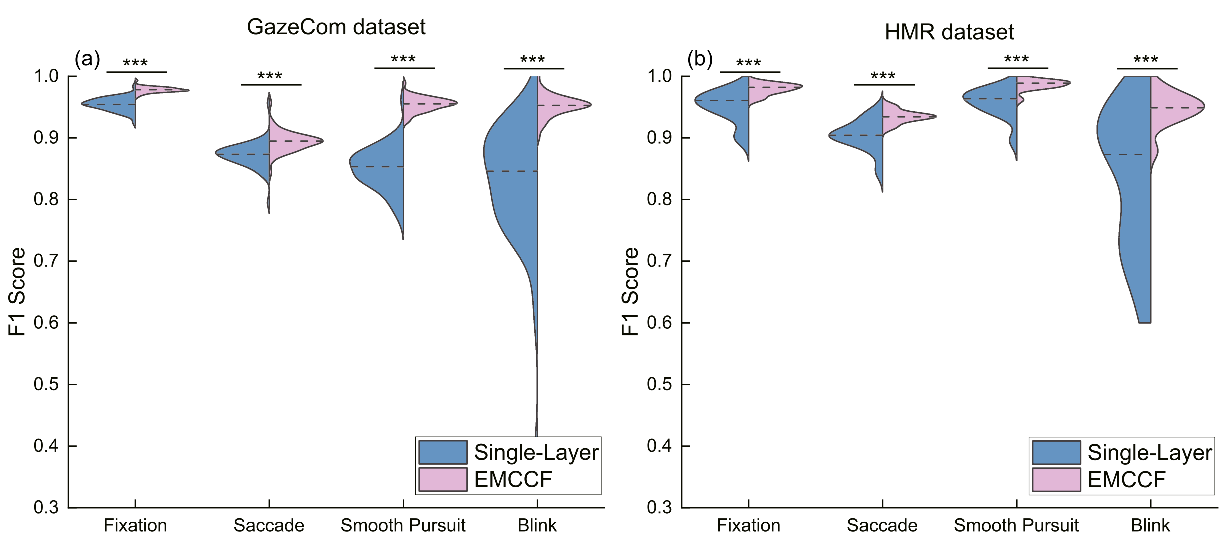

Fig. 11. The comparisons of F1 scores between the single-layer ensemble method and the proposed EMCCF method on the (a) GazeCom and (b) HMR datasets.

is evident that under the same hardware conditions, EMCCF required significantly less time for training. Similarly, during the testing phase, EMCCF also demonstrated a significant time advantage.

F. Effectiveness Comparison of EMCCF With a Single-Layer Ensemble Method

To compare the layered ensemble architecture with the traditional single-layer ensemble method, the number of layers was set to one to generate a single-layer ensemble. The configuration of base classifiers used in this single-layer ensemble method was kept consistent with those in the EMCCF model. The comparisons of F1 scores between the single-layer ensemble method and the proposed EMCCF method on both the GazeCom and HMR datasets are shown in Fig. 11. Statistical analysis indicated that the proposed EMCCF model achieved superior performance compared with the single-layer ensemble method (one-tailed paired t test, $p < 0.001$).

IV. Dissicussion

In this study, precision, recall, and F1-score were calculated for the GazeCom and HMR datasets, demonstrating that the proposed EMCCF model achieved a performance enhancement in eye movement classification compared with DL-based models. Compared with DL-based models, the EMCCF model was able to more effectively handle the data scarcity and class imbalance problems in eye movement classification. Additionally, the EMCCF model exhibited more robust performance across different participants when comparing classification accuracy among various models. Several potential reasons for the EMCCF model's superiority are hypothesized below.

First, in the EMCCF model, each layer's input combines the output of the previous layer with the original input, allowing the model to accumulate and utilize information from previous layers while effectively retaining information from the original input. Second, each layer of the EMCCF model incorporates four different types of base classifiers that combine both bagging and boosting ensemble learning methods, enhancing diversity among base classifiers. Furthermore, the combination of outputs from the previous layer with the original input increases data diversity. Because the effectiveness of ensemble learning partially relies on diversity among the base classifiers and the data [14], [26], the classification accuracy of the EMCCF model is enhanced. Third, unlike traditional DL models with a fixed number of layers [14], the EMCCF model employs an adaptive method to determine its number of layers. During training, each time a new layer is added, the model's overall performance is evaluated on a validation set. If the addition of a new layer does not result in any improvement in classification accuracy, the process of adding further layers is terminated. This adaptive layer growth strategy optimizes the model's generalization capabilities while avoiding unnecessary increases in computational complexity [14].

Moreover, the data presented in Fig. 9 show that the EMCCF model demonstrated advantages over DL models even with a limited number of multi-scale temporal windows, consistently maintaining a performance advantage. This finding indicates the EMCCF model's proficiency in extracting key information from limited features. Additionally, the results indicated that DL-based classification methods yielded high F1 scores for the fixation class but scores of zero for other classes when the number of time windows was minimal. This result occurred because all samples were identified as fixation. Furthermore, given the higher proportion of smooth pursuit in the HMR dataset compared with the GazeCom dataset, the results in Fig. 5 indicate that DL-based methods performed better in smooth pursuit classification with the HMR dataset compared with the GazeCom dataset. This indicates that the class imbalance problem was more effectively mitigated in the HMR dataset. However, the consistent performance of EMCCF across all classes in both datasets suggests that EMCCF is indeed capable of effectively addressing the class imbalance problem. The DL models, designed for processing time series, require longer time windows to capture long-term dependencies and enhance accuracy [30]. If the number of windows is limited, the model may fail to capture sufficient dynamic changes, leading to predictions that are biased towards the statistically most common class [31], namely the fixation class. Moreover, the EMCCF model combines both bagging and boosting ensemble learning methods to harness the advantage of diversity intrinsic to ensemble learning. This combination enables the EMCCF model to reduce reliance on extensive data samples. By combining diverse base classifiers, the model enhances its generalization capacity and stability across unseen data, thereby mitigating potential performance volatility or overfitting issues caused by insufficient data [28], [32].

Furthermore, computational complexity and scalability are crucial for evaluating the EMCCF model. Fig. 10 demonstrates that EMCCF exhibited significantly lower computational complexity in both the training and testing phases compared with the baseline models, which indicates that the proposed EMCCF method has advantages in terms of computational complexity. Furthermore, although the GazeCom dataset is nearly three times the size of the HMR dataset, the running time of the EMCCF model shows a trend of linear growth rather than explosive growth, which demonstrates that the proposed EMCCF model has scalability in handling large datasets.

The results in Fig. 11 show that the signal-layer ensemble method without layered ensemble architecture was inferior to the proposed EMCCF, which suggests that the layered ensemble architecture of EMCCF can be helpful for improving the model classification performance.

V. Conclusion

In the current study, we introduced the novel EMCCF method. The experimental results indicated that the EMCCF method surpassed existing DL-based eye movement classification methods in terms of performance and demonstrated enhanced robustness across various participants. Moreover, EMCCF reduced the dependency on large datasets, thereby augmenting the model's generalization performance when encountering new data. These attributes suggest that the EMCCF has considerable promise for eye-tracking tasks that demand good performance with limited data availability, particularly in cross-participant studies.

Appendix A
Maximum Number of Windows Determination

Similar to the studies of the studies of Startsev et al. [11] and Elmadjian et al. [13], pre-analysis was executed before determining the number of multi-scale time windows in this paper (Fig. 12). The pre-analysis revealed that, when the number of windows was set to 7, the performance of all models reached a relatively high level. However, beyond this point, there was no significant improvement of performance. Therefore, setting

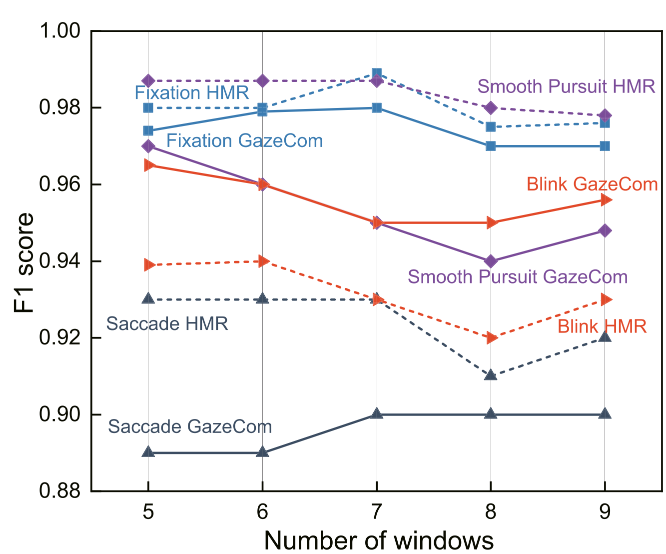

Fig. 12. Pre-analysis, the number of windows is selected from 5 to 9.

the maximum number of multi-scale time windows to 7 was a reasonable choice to balance efficiency and accuracy.

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